A Deterministic Judge That Cannot See Its Own Evidence

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Abstract

We evaluate a deterministic, environment-grounded judge that checks whether an LLM's explanation cites the sensors a graph explainer actually selected — no LLM judges another LLM. On 93 traffic-forecasting scenarios it works exactly as designed, yet cannot distinguish a real explanation from one whose evidence is a seeded uniform random draw: precision and hallucination match to four decimals. We trace this to a near-flat, unstable mask, show a downstream decision agent is equally indifferent, and report a failed entropy gate. A corrector built to raise grounding erases the one pre-loop measurement that did separate real evidence from fake.

1 Introduction

A spatial-temporal graph neural network forecasts speed at a traffic sensor; a mask-based explainer ranks which other sensors drove that forecast; an LLM turns the ranking into prose naming locations; a deterministic checker verifies those locations appear in the ranking. The pipeline is *explainer* $\rightarrow$ *LLM* $\rightarrow$ *verifier* $\rightarrow$ *decision*. The verifier is a decidable function of the explanation artifact and a node table, so it is immune by construction to the position, verbosity and self-enhancement biases afflicting LLM judges (Zheng et al., 2023). That immunity buys less than it appears to. The judge correctly scores citation–artifact correspondence, cannot tell whether the artifact carries any information, and a correction loop added to improve grounding destroys the one signal that would have revealed the evidence was fabricated. Our contribution is a set of controls — random and mismatched evidence, a second explainer, a decision-level replication, a failed detector — locating which part of a grounded evaluation stack is load-bearing.

2 Setup

The forecaster is a Graph WaveNet-style ST-GNN (Wu et al., 2019) on METR-LA (207 sensors); the explainer learns a soft node mask after GNNExplainer (Ying et al., 2019), and its top-k (k=8) entries are rendered into the prompt as the only admissible causes. The judge resolves each free-text citation to a node-id set via an escalating ladder (exact sensor, exact region, fuzzy, geographic), counting a hit when that set intersects the top-k; precision, recall, F_1 and hallucination ($1 -$ precision) follow. The resolver scores $29/30 = 96.7\%$ on hand-labeled cases — a **single-labeler validation on 30 cases**, not a reliability estimate. Five conditions share one 93-scenario corpus: **A** (full explanation), **B** (prediction only), **C** (no city context), **A_rand** (importances replaced by a seeded uniform draw) and **A_mismatch** (evidence from a different sensor's real explanation, via a seeded derangement). All use `llama3.1:8b`; every call is logged, and all non-LLM stages are seeded and byte-reproducible.

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Table 1: Faithfulness by condition. One 93-scenario corpus, epoch-34 checkpoint, `llama3.1:8b`; brackets are 95% bootstrap CIs (10,000 resamples, seed 42).

Condition	Precision	Recall	F_1	Hallucination
A (full explanation)	0.9946	0.5927	0.7246 [0.6930, 0.7553]	0.0054 [0.0000, 0.0161]
B (no explanation)	0.1720	0.0712	0.0896	0.8280
C (no city context)	0.9946	0.6048	0.7277	0.0054
A_rand (random evidence)	0.9946	0.5645	0.7013 [0.6681, 0.7334]	0.0054 [0.0000, 0.0161]
A_mismatch (wrong sensor)	0.9892	0.6008	0.7275 [0.6959, 0.7591]	0.0108 [0.0000, 0.0269]

3 Results

The judge is valid, and blind. A_rand replaces the explainer's importances with a seeded uniform draw and changes nothing else — same scenarios, same predicted speeds, same template, prompt lengths within 0.4% of A. It scores precision 0.9946 and hallucination 0.0054, *identical to A at four decimal places*, with paired differences of exactly 0.0000 on both and 95% CIs $[-0.0161, +0.0161]$ spanning zero. A_mismatch, shown a real explanation belonging to a different sensor, reaches F_1 0.7275 against A's 0.7246. Condition B makes this interpretable: deleting the explanation entirely does collapse the judge (precision $0.9946 \to 0.1720$, hallucination $0.0054 \to 0.8280$). The judge is not broken — it is valid for what it measures, correspondence between citations and a supplied artifact, and blind to whether that artifact means anything.

The mask is close to flat. The artifact carried little more information than noise to begin with. On an independent 207-target sweep of the same explainer, covering 80% of a target's off-self importance mass takes on average 146.6 ± 4.7 (congested) and 154.0 ± 2.9 (free-flow) of 206 available sources.[1] A ranking needing three quarters of the network for four fifths of its mass is close to flat, and the top-8 in every prompt is the head of that flat ranking.

And its top-k is close to unstable. Flatness predicts instability, and we measure it three ways. Splitting each target's 24 windows into two stratified halves and re-deriving the top-8 from each gives a Jaccard of 0.2256 over the 200 free-flow targets present in both halves and 0.1399 over the 28 congested ones, against a two-independent-random-draws reference of 0.0165 and 0.0182 respectively — above chance, but far from a determined object.[1] Raising the explainer's sparsity coefficient does not help: over 40 targets the split-half Jaccard is 0.2323 ± 0.1461 at the committed λ, 0.2385 ± 0.1739 at 3λ and 0.2287 ± 0.1831 at 10λ.[2] Re-running the explainer at a different random seed on 10 of those targets recovers only 0.3480, 0.3149 and 0.3015 of the top-8 at the three settings. Even the arithmetic is load-bearing: solving the same target and window at the same seed on CPU versus MPS agrees on importance *magnitudes* (Pearson 0.9193) while agreeing on top-8 *identity* only 0.3991 of the time.[3] Substituting a uniform draw for an object this weakly determined is a smaller intervention than it sounds.

A second explainer scores the same. On a 28-scenario subset where SHAP (KernelExplainer) was run beside the mask, the two methods' top-k sets overlap by a mean Jaccard of 0.0221 yet receive *identical* judge precision 0.9821 and hallucination 0.0179 (F_1 0.7423 vs 0.6918).[4]

The loop erases the one signal that existed. One measurement did separate real from fabricated evidence: before any correction, 64.5% of scenarios met the 0.70 grounding threshold on real explanations against 41.9% on random ones. A corrector — score the advisory, name the top-k nodes it failed to cite, re-prompt, up to three rounds — drives both to the ceiling: real reaches 100.0% at F_1 0.8738, random 98.9% at F_1 0.8903, the fabricated condition finishing marginally *higher* (Figure 1). **The loop built to make the system more reliable is precisely what destroys the only**

[1]Run `20260824T235016Z__influence_graph_solves`, `metrics.json`.

[2]Run `20260825T201433Z__grounding_without_information`, `part4_metrics.json`, `settings[]`. Seed-repeat and entropy figures below come from the same file.

[3]Run `20260824T235643Z__device_divergence`, `device_divergence.json`; 3 cases, Spearman rank 0.5778, top-20 Jaccard 0.3576.

[4]`shap_comparison_summary.json`, `aggregate.{A,D}` and `gnn_vs_shap_topk_overlap`; n=28.

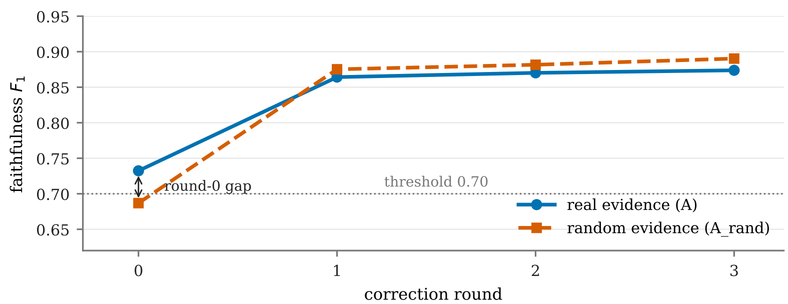

Figure 1: Faithfulness F_1 by correction round, real evidence (A) versus random evidence (A_rand), n=93 each, threshold 0.70. The arms are separated before any correction and indistinguishable after one round; by round 3 the fabricated condition is marginally higher. Series differ in colour, dash and marker so the figure survives greyscale.

measurement that revealed the evidence was fake. What survives is agreement between prose and artifact, which the loop optimises directly; what does not is the diagnostic value of failing to agree.

3.1 The blindness reaches the decision, not just the score

A judge that cannot see its evidence would matter less if a downstream agent could. We ran the decision simulation on 150 stratified scenarios $\times$ 3 seeds, asking an agent to choose one of five traffic interventions, scored against a model-in-the-loop ground truth. Both arms are present for every scenario and seed, so the contrast is paired *within* scenario and does not involve any baseline. Showing the agent the real explanation rather than a random one changes accuracy by $+0.0200$ (95% CI $[-0.0289, +0.0711]$) and delay reduction by $+0.1153$ mph (95% CI $[-1.3158, +1.5452]$), both bootstrapped over 10,000 resamples of scenarios.[5] **Both intervals span zero.** At the scenario level the real explanation wins 32 and loses 26 on accuracy with 92 ties, and wins 54 and loses 48 on delay reduction with 48 ties. We deliberately do not frame this against the no-explanation baseline: that baseline is unstable across sample sizes — 0.2756 accuracy on this 150-scenario subset against 0.2327 on the full 444 — by more than either explanation arm exceeds it on either, so an above-baseline test would report the sampling, not the effect.

3.2 A cheap detector that did not replicate

If the top-k is unstable because the mask is flat, then mask entropy should predict instability, and entropy is computable from a single solve whereas measuring stability costs a second one. We pre-registered that test before running it: normalised Shannon entropy of the off-self mask, correlated against the same split-half top-8 Jaccard, with success defined in advance as Spearman $\rho < -0.30$ *and* a bootstrap CI excluding zero, plus a threshold chosen on the sweep alone flagging all 12 committed explanations. Pooled over 120 (target, setting) pairs, $\rho = -0.2008$ with 95% CI $[-0.4229, +0.0452]$ for the entropy of the window-averaged mask and $\rho = -0.1732$, CI $[-0.3938, +0.0513]$ for the mean per-window entropy — the second being the only form obtainable from a single solve, and therefore the only one that would have made this cheap. Both intervals include zero; the threshold flagged 3 of 11 distinct committed explanations, not all of them.[6] **Both pre-registered criteria are falsified and we report them as such.** A single-setting version of this test at n=40 did appear to pass; it did not survive the full pool, which is the ordinary fate of an underpowered result and the reason the criteria were fixed in advance.

[5] `paired_decision_ab.json` in the grounding run's processed directory, computed from `part3_per_decision.csv`. Resampling unit is the scenario, not the decision, because a scenario's three seeds share a window, a ground-truth action and an explanation.

[6] `verdicts.json`, P4; both sub-verdicts recorded FALSIFIED. The 12 committed explanation files cover 11 distinct (target, window) pairs.

4 Facet mapping

A deterministic checker's coverage of judge-quality facets is unusually easy to state, because most of it is settled by construction rather than by measurement. *Surface-form sensitivity*, *positional bias*, *sycophancy* and *inter-judge inconsistency* are all removed outright: there is no judge model to exhibit them, the same input yields the same verdict on every run, and rerunning the scorer changes nothing. *Verbosity bias* is likewise absent: the score depends on the citation array, not answer length. Against that, three facets are untouched. *Criteria drift* is not addressed at all: the evaluation field is self-declared by the same project that built the system being evaluated, with no external specification and no community standard. *Reasoning-chain validity* is not addressed: the judge checks which locations are cited, never whether the causal story told about them is coherent or correct, so a reversed edge or an invented propagation order scores clean. And *construct validity* is exactly what this paper puts in question: a judge scoring random evidence identically to real evidence is measuring citation compliance, not grounding. A filled disclosure template is provided in Appendix A.

5 Limitations

(a) Faithfulness here is agreement with what the model was *shown*, not with what is *true*; there is no ground-truth causal attribution for a real road network, which is why A_rand is diagnostic at all. **(b)** Condition B differs from A in several ways at once (no importance table, no propagation path, different retrieved context), so we make no mechanistic claim about which absence causes its collapse. **(c)** One claim type (citation membership), a self-declared evaluation field, and a single LLM family per run: every condition in Table 1 used `llama3.1:8b`, and we make no cross-model claim — a second model appears only in a separate cross-city study that is excluded here as a different corpus. **(d)** Conditions A/B/C and A_rand/A_mismatch were scored by different versions of the entity resolver (fixed 2026-07-21, between runs). This does not affect the precision/hallucination results above — precision is identical to four decimal places between A and A_rand — since a resolver-version change can only affect which citations are matched, not whether a match is correct. It may affect the smaller recall/F_1 differences between conditions; a same-corpus proxy bounds the resolver's own contribution at approximately 0.005–0.014 F_1. We restrict our central claim to the precision/hallucination axis, which is unaffected. **(e)** The resolver was validated on 30 hand-labeled cases by a *single labeler*; there is no second annotator and no inter-annotator agreement, so 96.7% is a development check, not a measured reliability. **(f)** The supporting measurements carry small samples of their own: the seed-repeat Jaccard rests on 10 targets, the device comparison on 3 cases, the SHAP comparison on 28 scenarios, and the congested split-half on 28 targets. We report them as indicative, and none of them carries the central claim, which rests on Table 1. **(g)** The decision simulation's ground truth is model-in-the-loop — what the trained forecaster believes an intervention would do, not an observed outcome — so it inherits that model's biases and is internally consistent rather than externally valid.

6 Related work

LLM-as-judge biases are set out by Zheng et al. (2023). Citation and attribution evaluation are formalised by ALCE (Gao et al., 2023) and AIS (Rashkin et al., 2023), which separates attribution from correctness by definition; our results instantiate that separation where no gold reference exists. Shi et al. (2023) show irrelevant context degrades task accuracy — here the irrelevant content is the evidence the model is told to cite, and the measured quantity is compliance.

The stability results sit on an existing line of work on GNN-explainer evaluation. Sanchez-Lengeling et al. (2020) benchmark attribution against computable ground truth on synthetic graphs; our setting has none, which is what makes a randomised-evidence control necessary rather than optional. Faber et al. (2021) show that comparing explanations to ground truth is itself unsound when the GNN does not use the ground-truth edges, separating responsible, causing and explanatory motifs; our mismatch is one step further downstream, between an explanation and a judge that consumes it. Agarwal et al. (2023) provide generators and a library for evaluating explanation quality. Our numbers agree with that literature; what we add is the downstream consequence — an unstable, near-flat attribution is indistinguishable from noise to the verifier and to the agent built on it.

To our knowledge this is the first evaluation where the cited source is a *GNN-explanation artifact*, checked by a *deterministic* verifier with no gold reference, paired with a *correction loop* whose effect on the noise signal is measured.[7]

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A JUDGe Judge Deployment Disclosure Template (filled)

Judge identity. Deterministic program, not a model. Entity resolver plus set-intersection scoring over the explainer's top-k. No LLM is used to judge another LLM's output. Version-sensitive: the resolver was changed 2026-07-21.

What is judged. A single claim type: whether a free-text location in the LLM's `cited_causes[]` array resolves to a node set intersecting the explainer's top-k. The LLM's prose is not read. A refusal (no citations) scores hallucination 1.0, identical to confidently inventing causes.

Scale and outputs. Precision, recall, F_1, hallucination rate $(1 - \text{precision})$, per advisory, averaged over 93 scenarios.

Ground truth. None available. The reference is the explainer's own top-k, so the judge measures correspondence, not correctness.

Validation. Resolver: $29/30 = 96.7\%$ on hand-labeled cases, one labeler, no inter-annotator agreement, threshold set at 90% in advance.

Known blind spots. Cannot distinguish real from random evidence (Table 1); cannot distinguish overreach from falsehood, since both score as 1 – precision; does not read the reasoning chain; scores a refusal identically to a fabrication.

Criteria provenance. Self-declared by the authors of the system being evaluated. Not externally specified, not community-standardised.

[7] C_RICH (fuller context) and D_CONTRA (contradictory context) ran on this corpus in a separate LLM draw and did not move hallucination materially; being a different draw they are omitted from Table 1.

Reproducibility. All non-LLM stages seeded and byte-reproducible. LLM calls are logged in full but are *not* byte-reproducible: they run at temperature 0.1 on an unseeded sampling path, chosen so results remain comparable with previously committed numbers produced on that path.

B Claim-to-source map

Every number in the main text traces to one of the following. Run directories are under `evaluation/results/`; `GWI` abbreviates run `20260825T201433Z__grounding_without_information`.

Claim	Value	Source
A / B / C rows	Table 1	`claims.csv` ← `faithfulness_summary.json`
A_rand / A_mismatch rows	Table 1	`claims.csv` ← GWI `part1_summary.json`
Bootstrap CIs, paired diffs	10k, seed 42	GWI `part1_summary.json`, `bootstrap_ci`
Corpus, epoch 34, n=93	2026-07-05/06	`claims_provenance.json`, `corpus`
Prompt lengths within 0.4%	+0.18%, +0.38%	GWI `part1_token_parity.json`
80% mass of importance	146.6±4.7 / 154.0±2.9 of 206	`20260824T235016Z__influence_graph_solves`, `metrics.json`
Split-half Jaccard, 207-target	0.2256 / 0.1399; ref 0.0165 / 0.0182	same file, `per_regime.*.stability`
Split-half Jaccard, 3 λ	0.2323 / 0.2385 / 0.2287	GWI `part4_metrics.json`, `settings[]`
Seed-repeat Jaccard, 3 λ	0.3480 / 0.3149 / 0.3015 (n=10)	same file
CPU vs MPS divergence	Pearson 0.9193, top-8 J 0.3991	`20260824T235643Z__device_divergence`, `device_divergence.json`
SHAP overlap and scores	J 0.0221; F_1 0.7423 / 0.6918	`shap_comparison_summary.json`
Round-0 grounding	64.5% real / 41.9% random	`active_grounding_summary.json`; GWI `part2_summary.json`
Loop curve, Figure 1	real F_1 0.7323→0.8738; random 0.6868→0.8903	same two files, `per_round[]`
Loop reach endpoints	100.0% / 98.9%	same two files
Paired decision contrast	$+0.0200$; $+0.1153$ mph	GWI `paired_decision_ab.json`
Baseline instability	0.2756 (n=150) vs 0.2327 (n=444)	same file, `raw_baseline_instability`
Entropy gate ρ and CIs	-0.2008, -0.1732; 3/11 flagged	GWI `verdicts.json`, `P4`
Resolver 29/30 = 96.7%	threshold 0.90	`resolver_accuracy.json`
Resolver proxy 0.005–0.014 F_1	same-corpus bound	`claims_provenance.json`